ISWC 2026 · HALF-DAY TUTORIAL

Façade-X: Querying Any Format as a Knowledge Graph

Integrate CSV, JSON, XML, spreadsheets and more using SPARQL alone.

📍 Bari, Italy · 📅 25 October 2026 · ⏱️ 14:10–18:00

SPARQL | Knowledge Graphs | Data Integration | Hands-on

The afternoon in one line

Any data format → a knowledge graph → SPARQL alone.

- The Façade-X idea and its foundations
- Hands-on with SPARQL Anything
- (coffee break)
- How it is used in practice, taught, and standardised

Agenda

Time	Session	Who
14:10–14:50	**Slot 1 · The Problem and the Idea**	
14:10–14:30	Introduction to Façade-X	Enrico
14:30–14:50	Foundations: towards a theory of façade-based data access	Luigi
14:50–16:10	**Slot 2 · Hands-on with SPARQL Anything**	
14:50–15:20	Live querying of CSV, JSON and XML	Luigi
15:20–15:45	Functions and magic properties	Luigi
15:45–16:10	Guided exercises with real datasets	Enrico
16:10–16:30	**Break · SPARQL Anything Clinic — bring your data**	**All**
16:30–18:00	**Slot 3 · From Practice to Standards**	
16:30–17:30	Industry talks: railway KG · TriplyDB deployment · Public Procurement Data Space	Mathias, Martin, Ivo
17:30–17:40	Teaching KG management with SPARQL Anything	Ryan
17:40–18:00	W3C Data Façades CG: roadmap and discussion	Enrico

Presenters

Luigi Asprino
Università Telematica San Raffaele, Roma (IT)

Co-founder of SPARQL Anything · chair, W3C Data Façades CG

Enrico Daga
KMi, The Open University (UK)

Co-founder of SPARQL Anything · chair, W3C Data Façades CG

Martin van Harmelen
Triply (NL)

SPARQL engine design and ELT pipelines on TriplyDB

Mathias Vanden Auweele
MATDATA (BE)

Railway infrastructure data and production KGs

Ryan Shaw
UNC Chapel Hill (US)

Teaches KG management with SPARQL Anything

Ivo Velitchkov
Kvistgaard (DK)

Consultant and trainer in semantic technologies

Show of hands ✋

Raise your hand if you…

1. have **heard of** Façade-X or SPARQL Anything before today
2. have **tried** SPARQL Anything, even once
3. **use** it in your work: research, a project, a product, teaching
4. have built a knowledge graph from CSV, JSON or XML **some other way**: RML/YARRRML, SPARQL Generate, scripts…

And: what data gives you the most trouble? Keep it for the clinic.

What you will learn today

IN PRACTICE

- Query **CSV, JSON, XML, Markdown, APIs** as a knowledge graph, with SPARQL alone
- Map any source to **your own vocabulary** in one `CONSTRUCT`
- **Join formats** in a single query
- Turn queries into **pipelines**: templates, parameters, reuse
- Use **SPARQL Anything** from the command line and from Python

BEYOND PRACTICE

- The **Façade-X idea**: one minimal model (containers, slots, values) for any format
- Its **theory**: what we can know about a query before touching the data
- How it is **deployed** in production: railways, procurement, TriplyDB
- How it is **taught** in KG courses
- How it is becoming a **standard**, and how to join the W3C CG

Before we start

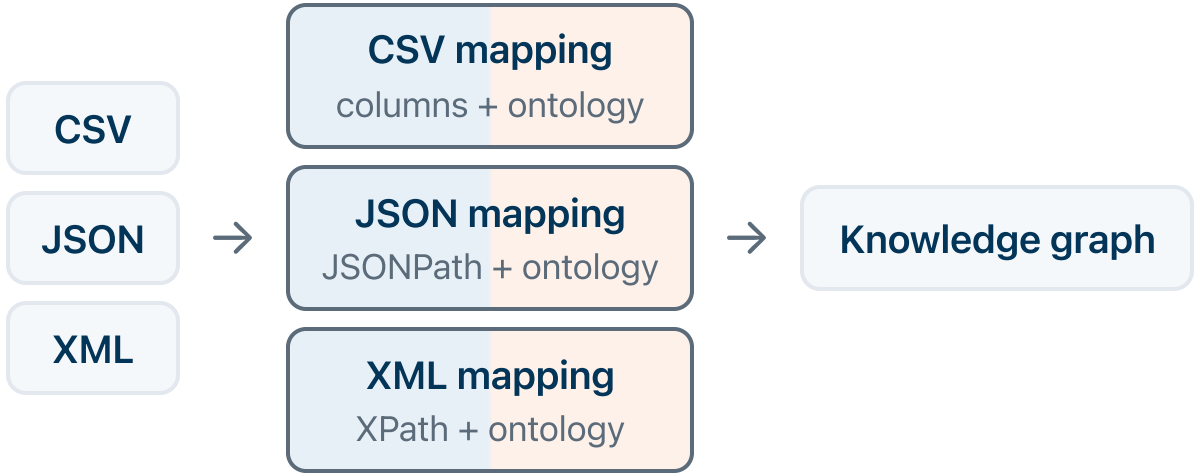
- **SPARQL Anything** downloaded, or **PySPARQL Anything** in Python / Jupyter
- No local setup? A **Google Colab** notebook is available
- Slides, notebooks and exercises: **w3c-facade-x.github.io/iswc2026-tutorial**
- Questions any time — and keep your own data for the clinic during the coffee break.

SLOT 1 · THE PROBLEM AND THE IDEA

Let's start.

w3c-facade-x.github.io/iswc2026-tutorial

Building a knowledge graph: the usual way



Re-engineering and **re-modelling** in the same artefact. Why is it a problem?

One mapping per format

New format, or a format change: rewrite the mapping.

Two skills, one artefact

Format expert and domain expert must edit the same rules.

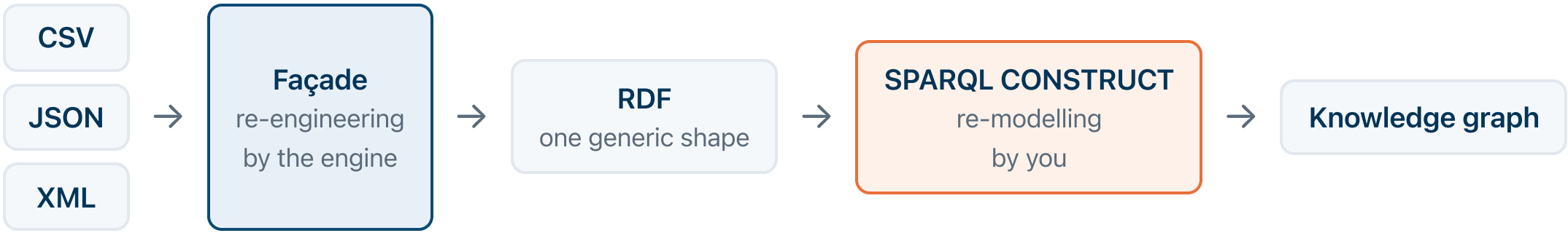
A second language

RML, R2RML, XSPARQL… next to SPARQL, with their own tools.

Hard to maintain

Change the ontology: touch format code. Change the format: touch the ontology.

Building a knowledge graph: with Façade-X



Formats handled once

In the engine, for everyone. No format code in your mapping.

One skill per step

You write only the re-modelling: your domain, your ontology.

SPARQL only

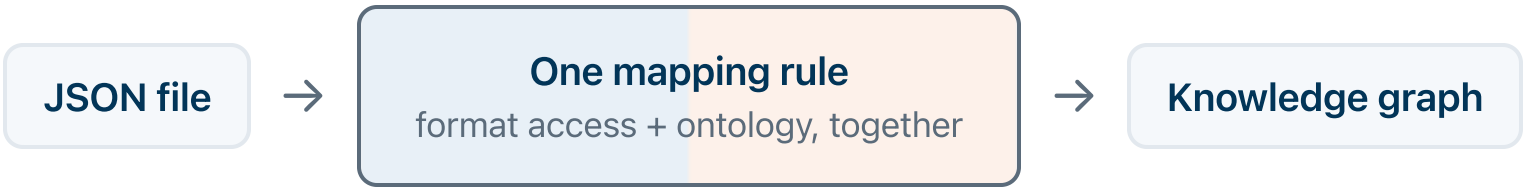
The language you already know, with all its features.

Easy to maintain

Change the source: keep the mapping. Change the ontology: keep the source.

Daga et al., SEMANTiCS 2021 · Asprino et al., ACM TOIT 2023

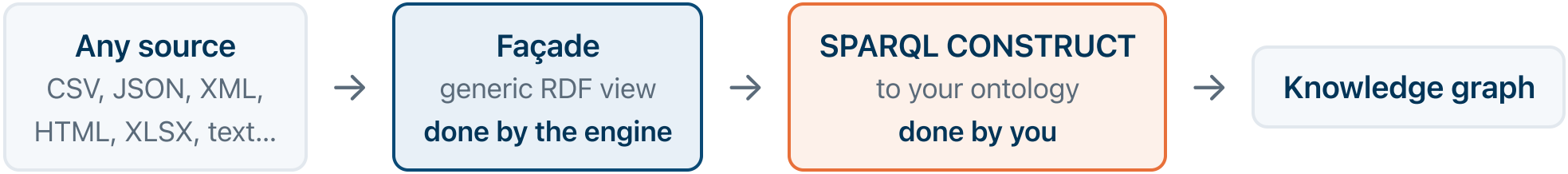
Most approaches merge them




```
mappings:
  artist:
    sources:
      - [artists.json~jsonpath, '$[*]']
    s: ex:$(name)
    po:
      - [a, schema:Person]
      - [schema:name, $(name)]
```

Same for R2RML, RML, XSPARQL, scripts. **New format → new rule.** And a second language next to SPARQL.

Façade-X: keep them apart



Re-engineering → the façade

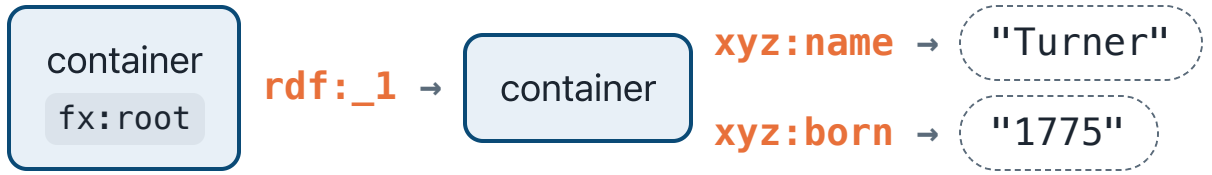
One generic structure for every format. No format knowledge in your queries.

Re-modelling → SPARQL

The language you already know. Joins, filters and functions come for free.

Daga, Asprino, Mulholland, Gangemi. Facade-X: an opinionated approach to SPARQL Anything. SEMANTiCS 2021.

The meta-model: containers, slots, values



Container

A file, a row, an object, an array, an element.

Slot · by number

`rdf:_1`, `rdf:_2` … order: rows, array items, children.

Slot · by name

`xyz:name` … keys, headers, attributes.

Value

A literal. A container may also have a **type**.

Same data, same façade

CSV

```
name,born
Turner,1775
```

```
[] a fx:root ;
  rdf:_1 [
    xyz:name "Turner" ;
    xyz:born "1775" ] .
```

JSON

```
[ {"name": "Turner",
   "born": 1775} ]
```

```
[] a fx:root ;
  rdf:_1 [
    xyz:name "Turner" ;
    xyz:born "1775"^^xsd:decimal ] .
```

XML

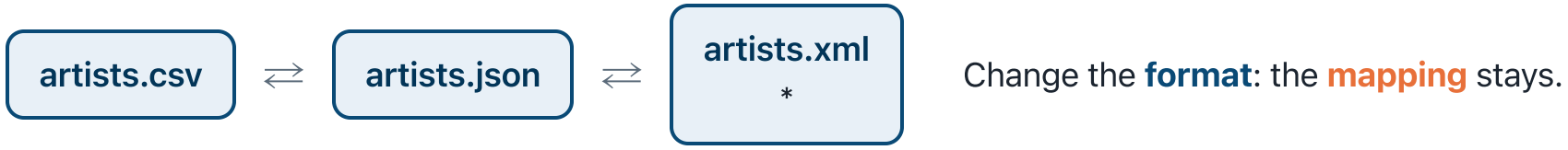
```
<artists>
  <artist name="Turner" born="1775"/>
</artists>
```

```
[] a fx:root, xyz:artists ;
  rdf:_1 [ a xyz:artist ;
    xyz:name "Turner" ;
    xyz:born "1775" ] .
```

Three formats, one shape. The highlighted parts are the only differences.

One query, both jobs

```
CONSTRUCT {
  ?artist a schema:Person ; schema:name ?name ; schema:birthDate ?born .
} WHERE {
  SERVICE <x-sparql-anything:location=artists.csv,csv.headers=true> {
    ?row xyz:name ?name ; xyz:born ?born .
  }
  BIND (fx:entity(ex:, ?name) AS ?artist)
}
```



Change the **format**: the **mapping** stays.

Why "opinionated"

Minimal

Four notions. Learn one model, not one per format.

SPARQL only

No second language. All of SPARQL is available.

Virtual or materialised

Query sources directly, or save the result as RDF.

The price

Some format details are left out, by design.

Read more

The idea

Daga, E., Asprino, L., Mulholland, P., Gangemi, A.
Facade-X: an opinionated approach to SPARQL Anything.
SEMANTiCS 2021. Studies on the Semantic Web 53: 58–73.

Introduces the Façade-X meta-model and SPARQL Anything.

The method

Asprino, L., Daga, E., Gangemi, A., Mulholland, P.
Knowledge graph construction with a façade: a unified method to access heterogeneous data sources on the web.
ACM Transactions on Internet Technology 23(1): 1–31, 2023.

Formalises Façade-X and shows its expressive power: demonstrates it can represent any format defined by a context-free grammar.

Next, the theory: Asprino & Daga, *Towards a theory of Façade-X data access*, TGDK 2026.

What comes next

